

(Pre-)registration template for experience sampling methodology (ESM) research

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Kirtley, O. J., Lافit, G., Achterhof, R., Hiekkaranta, A. P., & Myin-Germeys, I. (2020, March 12). A template and tutorial for (pre-)registration of studies using Experience Sampling Methods (ESM). <https://doi.org/10.17605/OSF.IO/2CHMU>

The template is accompanied by an open access tutorial paper: Kirtley, O. J., Lافit, G., Achterhof, R., Hiekkaranta, A. P., & Myin-Germeys, I. (2021). Making the Black Box Transparent: A Template and Tutorial for Registration of Studies Using Experience-Sampling Methods. *Advances in Methods and Practices in Psychological Science*. <https://doi.org/10.1177/2515245920924686>

This template is adapted from the Preregistration Challenge template (Mellor, D. T., Esposito, J., Hardwicke, T. E., Nosek, B. A., Cohoon, J., Soderberg, C. K., ... Speidel, R. (2019, February 6). Preregistration Challenge: Plan, Test, Discover. Retrieved from osf.io/x5w7h). We have also added sections (sections 10, 19, and 21) that were adapted from the Preregistration of Secondary Data Analysis template of Van den Akker, O., Weston, S. J., Campbell, L., Chopik, W. J., Damian, R. I., Davis-Kean, P., ... Bakker, M. (2019, November 20). Preregistration of secondary data analysis: A template and tutorial. <https://doi.org/10.31234/osf.io/hvfmr>

How to register your study on the OSF using this template:

- Download and complete the Word/PDF version of this template
- When you are ready to register your study on the OSF, go to the “registrations” tab within the relevant OSF project and create a new registration.
- When presented with the different options for creating a new registration, select “open-ended” registration.
- After completing the “registration meta-data” form, you will be directed to the “summary” page. Cut and paste the completed ESM registration template document into this “summary” box.
- If you have code or equations within your registration document, it is best to also attach a PDF copy of the registration via the “add supplemental files or additional information” option, because equations/code do not render well when copy and pasting. Code files can also be attached as supplementary materials for the registration, e.g. as R or Stata files.
- You then have the chance to review your registration before hitting the “register” button and completing the final steps.
- Happy registration!

Study Information

Title

The importance of emotion characteristics for emotion regulation dynamics in daily life

Authors (required)

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Description (optional)

In this study, we examine how emotional characteristics are related to emotion regulation (ER) strategy use in daily life, separately for positive and negative emotions, using intensive longitudinal data from a 60-day ESM study.

For negative emotions, we examine two emotional characteristics, namely emotional intensity and emotional control. For ER strategies use, we examine six ER strategies, namely distraction, rumination, suppression, relaxation, expression, and reappraisal.

For positive emotions, we examine emotional intensity as emotional characteristic (emotional control for positive emotions was not assessed). For ER strategy use, we examine two ER strategies, namely expression and savoring.

Relationships will be assessed at the within-person level, considering both linear and quadratic relationships.

Hypotheses (required)

Our hypotheses are based on earlier research (e.g., Blanke et al., 2022; De France & Hollenstein, 2022; Lennarz et al., 2019; Medland et al., 2020; O'Toole et al., 2017). Given the number of ER strategies assessed, we refrained from forming specific hypotheses for every single ER strategy, but rather aimed to systematically study the relationship between emotional characteristics and a broad range of ER strategies.

H1: Linear effects for emotional intensity and ER strategy use: In general, we expect a positive relation between emotional intensity and ER strategy use, both for positive and negative emotions (directional), however we did not form any hypotheses regarding for which ER strategy this relationship would be strongest.

H2: Linear effects for emotional controllability and ER strategy use: We expect a relation between negative emotional control and ER strategy use, but due to limited and mixed previous research, we did not form specific hypotheses about the direction of effects (non-directional). Note that we did not assess positive emotional control.

H3: Quadratic effects for emotional characteristics and ER strategy use: In addition to linear effects, we will also test for quadratic effects. Here, we hypothesize that the within-person relation between emotional characteristics and ER strategy use may be quadratic, where ER strategies are especially used at medium emotional intensity/control, but less at low emotional intensity/control and high emotional intensity/control (directional). Because we are not aware of any study that has addressed this question, we have no prior hypotheses on for which ER strategies quadratic effects would be most likely (non-directional).

Design Plan

Study type (required)

Observational study combined with an experimental within-person design (i.e., personalized micro-interventions based on individual affective responses). The present study only makes use of the observational data and controls for presence of micro-interventions.

Blinding (required for intervention studies)

Blinding is within-person. Participants did not know when the intervention is timed on their affect scores or at a random point in time. Personnel who interacted directly with the participants were not aware of which intervention was based on affect dynamics and which was random.

Is there any additional blinding in this study?

No

Study design (required)

Within-participants observational and experimental design

Participants were followed in daily life (observational part), plus micro-interventions were offered based on participants' individual affective stability (experimental design part; micro-interventions were applied during half of instabilities). For more information see 19. Manipulated variables.

Sampling Plan

Existing data (required)

Registration following analysis of the data (Data cleaning has already been done; see prior knowledge of data for further information on this).

Explanation of existing data (if applicable)

Name and briefly describe the data set(s), and if applicable, the subset(s) of the data you plan to use.

The current study will use a subset of variables (see variables section for names) from the Track your Mood Study, using data from all participants ($N=83$; but see section on Data exclusion for more information).

Specify whether this data is open or publicly available.

Data are not publicly available.

How can the data be accessed? Provide a persistent identifier or link if the data are available online, or give a description of how you obtained the dataset.

The first author (D. Maciejewski) is Principal Investigator of the Track your Mood-study and has direct access to the data.

Specify the date of download and/or access.

Data have already been accessed (because they were cleaned by the first author D Maciejewski). Data cleaning was finished on 28 October 2022.

If the data collection procedure is well documented, provide a link to that information. If the data collection procedure is not well documented, describe, to the best of your ability, how data were collected.

The protocol of the Track your Mood study can be found under <https://doi.org/10.17605/OSF.IO/FX3AY>.

Some studies offer codebooks to describe their data. If such a codebook is publicly available, link to it here or upload the document. If not, provide other available documentation. Also provide guidance on what parts of the codebook or other documentation are most relevant.

The codebook for the Track your Mood study is publicly available online at <https://osf.io/fx3ay/files/osfstorage/62344ceaac980a08aa9516fa>

ESM data collection procedure (required)

Note that the Track your Mood study had a Day questionnaire, an Evening questionnaire and micro-interventions. In this pre-registration, only data from the Evening questionnaire are used and as such only procedures relevant for this are described. Please see <https://doi.org/10.17605/OSF.IO/FX3AY> for the complete protocol of the “Track your Mood” Study.

Micro-interventions were never scheduled during the evening assessments. Nevertheless, we will control for presence of micro-interventions.

Study duration (number of days)

The ESM period was 60 days (+ onboarding day).

Type of sampling scheme:

The sampling scheme was signal contingent. Participants received one notification per day at 21:00 hours to fill in the evening questionnaire. The notifications occurred via their mobile phones through the application “m-path” [www.m-Path.io; Mestdagh et al. (2022)], which they had previously installed and set up. While 21:00 was set as a default notification time for all participants, individual adjustments could be made during the intake meeting with the researchers. Additional personalization of the participant schedules was also possible at request throughout the study weeks.

Total number and type of items (open-ended or closed) and estimated questionnaire duration. Indicate the total number of items per assessment, also when items are not used for the analyses specified in this pre-registration. A full list of items should be provided in an appendix.

Total number of items of the evening questionnaire was 28 items. Completion took around 2 minutes.

Time-out specifications. Here you can indicate details about the amount of time that participants have to respond to a questionnaire (i.e., response time), and the potential maximum amount of time that participants have to respond to one item, or complete the full questionnaire.

Following the initial notification, the ESM questionnaire was available to the participant for 2 hours. There was no additional time restriction once the questionnaire had been opened.

Additional passive monitoring.

None.

Incentivization and increasing participant engagement.

Participants could either receive a voucher (VVV or bol.com bon) for 25 Euros or 2.5 study participant points (applicable for e.g. psychology students). If participants dropped out, they would receive the proportional amount of money and/or points. The researchers contacted participants when compliance was lower than 70%. Compliance was checked every few weeks. Participants could not see their own compliance in the app.

Additionally, we offered participants to see their individual data after the study had been completed. Participants were encouraged to contact the research team with questions or issues via e-mail, phone call, or WhatsApp. The team was available via these mediums throughout the entire study period.

Please indicate any additional details that are relevant to the ESM data collection procedure.

See <https://doi.org/10.17605/OSF.IO/FX3AY> for additional information on the data collection procedure (recruitment, data-viewing sessions etc.)

Sample size (number of participants) (required)

We aimed at including 100 students for our study. The initial goal of 100 students was based on typical sample sizes of earlier studies (Dejonckheere et al., 2021; Eisele et al., 2021; Haines et al., 2016; Hjartarson et al., 2021; Medland et al., 2020; Mey et al., 2020). In total, 83 participants started the study, and 77 students completed the study fully (i.e., 6 participants dropped out during the 60 days).

Rationale for sample size: Temporal design and number of participants (if applicable)

See <https://doi.org/10.17605/OSF.IO/FX3AY> for a rationale on the sample size of the overall project.

For the current study, we conducted a power calculation with a Monte Carlo Simulation with 1000 replications using parameters estimates from other pre-existing data with code from the Shiny app from Lafit et al. (2021). Big thanks to Ginette Lafit, who adjusted the code, so that we could test for power of the quadratic effects! The code can be found in the file *Power-analyses_main.R*.

As pre-existing data, we used the publicly available data from the publication of Medland et al. (2020), which can be found under <https://osf.io/h4gx3>. We believe that this dataset is suitable for our power analyses for several reasons. First, the samples are comparable in that both are mostly composed of undergraduate students. Second, we used the same items for the ER strategies and for the emotional characteristics (with some adjustments occurring in the translation). The ER strategy items are based on the RESS-EMA, which was validated in the study of Medland et al. (2020). Note, that the original questionnaire had two items per strategy. For the Track your Mood study, we chose one item per scale. Please see <https://osf.io/vg2wq> for more information on which items we chose and why. For the power analyses, we are only using the items that we used in the Track your Mood study. Note that the time-scale differs, where the study by Medland et al. (2020) assessed ER strategy use 8 times per day for 7 days, and the Track your Mood study assessed ER strategy use 1 time per day for 60 days.

The following variables are available:

1. ER strategies (within-person level):

- Relaxation: dampening of autonomic arousal (we used item 2)
- Expression: active expression of emotion (we used item 1) (note that this is also referred to as engagement)
- Rumination: sustained attention (we used item 2)
- Reappraisal: cognitive reframing (we used item 1)
- Distraction: diverting attention (we used item 2)
- Suppression: inhibition of emotional expression (we used item 1)

2. Emotional characteristics (within-person level):

- emotion intensity
- emotion control

The results of our analyses can be found in the tables below. Overall, results indicate that power for the linear and quadratic effects depends on the emotional characteristic and ER strategy combination.

For most of the combinations, except for emotional intensity & distraction, emotional control & expression, emotional control & distraction, emotional control & rumination, there is sufficient power ($> .80$) for either the linear **or** the quadratic effect.

For emotional intensity, power is $> .75$ for the linear effect, but insufficient power for the quadratic term (.09 - .45).

For emotional control, power is not sufficient for any of the linear effects. However, there is sufficient power for the quadratic effect for relaxation, reappraisal and suppression ($> .82$).

Table emotional intensity:

predictor	Number of replications converged	power linear term	power quadratic term
relaxation	1000	1.00	0.18
expression	998	0.97	0.09
rumination	1000	1.00	0.18
reappraisal	998	0.95	0.32
distraction	999	0.75	0.29
suppression	1000	1.00	0.45

Table emotional control:

predictor	Number of replications converged	power linear term	power quadratic term
relaxation	1000	0.21	0.95
expression	999	0.19	0.18
rumination	1000	0.21	0.62
reappraisal	1000	0.71	0.82
distraction	1000	0.26	0.56
suppression	1000	0.06	0.93

Medland et al. (2020) have only data available on negative emotions. However, we do not expect that power will be much different for the case of positive emotions.

Stopping rule (if applicable)

N/A

Variables

Measured non-ESM/time invariant variables (if applicable)

We will use age in years and gender (male, female, non-binary) as a covariate, which are reported by participants at baseline.

The exact item wording and variable names can be found in the codebook, which is publicly available online at <https://osf.io/fx3ay/files/osfstorage/62344cea980a08aa9516fa>

Measured ESM/time-variant variables (required)

Here, we will report the English translations. The items in Dutch can be found in the codebook, which is publicly available online at <https://osf.io/fx3ay/files/osfstorage/62344cea980a08aa9516fa>

We assessed emotional characteristics and ER strategy use separately for positive and negative emotions. See the following figure for the study design and order of questions:

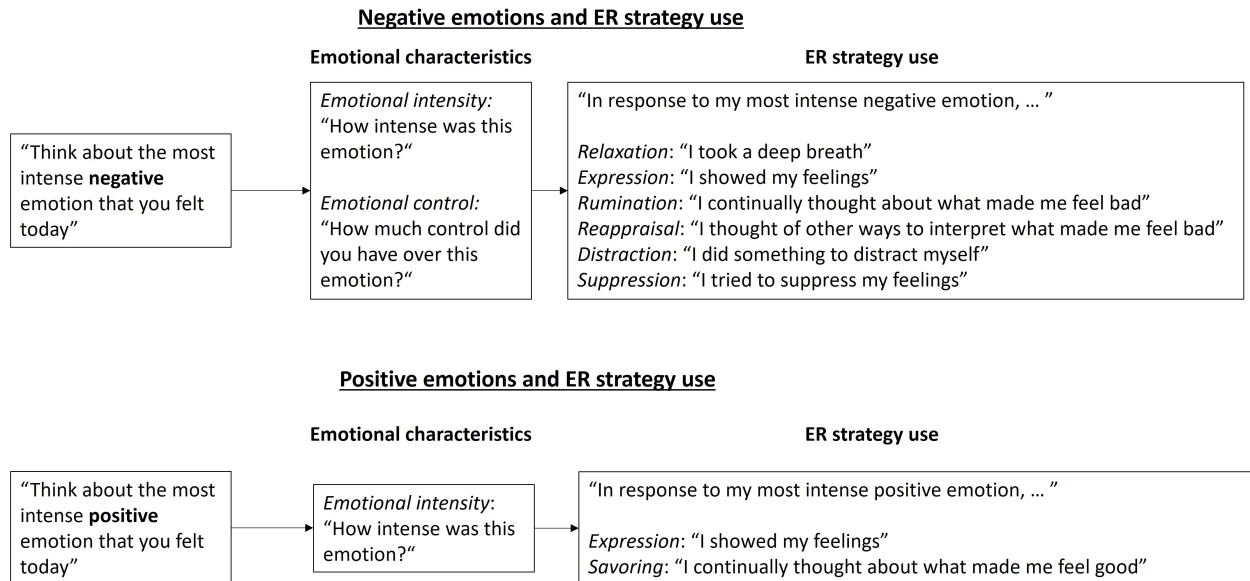


Figure 1: Figure 1. Design of measuring emotional characteristics and ER strategy use

The ESM predictor variables will be two measures of emotional characteristics for negative emotions: negative emotional intensity and negative emotional control (Medland et al., 2020). For positive emotions, we only assessed positive emotional intensity (and not controllability), because when we designed the study, we hypothesized that emotional control would play less of a role for positive emotions.

Participants were asked to think about the most negative/positive emotion of that day ("Think about the most negative/positive emotion that you felt today").

Emotional intensity was measured with the item "How intense was this emotion" with a VAS response scale ranging from 0 (*not at all*) to 100 (*very intense*).

Emotional control for negative emotions was measured with the item "How much control did you have over this emotion?" with a VAS response scale ranging from 0 (*not at all*) to 100 (*complete control*).

The ESM outcome variables will be ER strategies from the RESS-EMA questionnaire (Medland et al., 2020). The RESS-EMA only has items about the regulation of negative emotions. However, we based the

two positive ER strategy use items that we thought we most relevant to the maintenance of positive emotions on the RESS-EMA questionnaire (savoring and expression).

All items were rated on a VAS response scale ranging from 0 (*not at all*) to 100 (*very much*).

Participants were asked to indicate the intensity to which they used a certain ER strategy after being prompted with “In response to my emotions, . . .” The number of ER strategies assessed differed for negative versus positive emotions:

For regulation of negative emotions, six items were used to assess ER strategy use: “In response to my [most intense negative] emotion. . .”

Relaxation was measured with the item “I took a deep breath”.

Expression was measured with the item “I showed my feelings”.

Rumination was measured with the item “I continually thought about what made me feel bad”.

Reappraisal was measured with the item “I thought of other ways to interpret what made me feel bad”.

Distraction was measured with the item “I did something to distract myself”.

Suppression was measured with the item “I tried to suppress my feelings”.

For regulation of positive emotions, two items were used to assess ER strategy use: “In response to my [most intense positive] emotion. . .”

Expression was measured with the item “I showed my feelings”.

Savoring was measured with the item “I continually thought about what made me feel good”.

Please see <https://osf.io/fx3ay/files/osfstorage/62344ce61372e808bfcc59c2> for more information on translation of the RESS-EMA items.

Presence of micro-interventions will be coded as a time-varying dummy variable and will be included as a covariate (0 = participants received no micro-intervention that day, 1 = participant received a micro-intervention that day).

Open-ended questions (if applicable)

N/A for this study.

Indices (if applicable)

Gender (recoded): Because there are few males and only one non-binary person in our dataset, gender will be recoded for analyses into female (0) and other (1; including male and non-binary).

ER variability: We will calculate momentary ER variability in two steps. First, we calculate Bray-Curtis dissimilarity between the moment of interest and all other moments within the same participant. In total, there will be $(n - 1)$ comparisons, where n is the number of observations within a participant. Then, we take the mean of $(n - 1)$ instances of Bray-Curtis dissimilarity. In this way, momentary ER variability reflects how much a participant deviated from their usual style of emotion regulation at the moment of interest. This measure has been previously validated for the use in ESM research (see Lo et al., 2023). Given that this measure is new, the analyses using the ER variability index were exploratory (see under exploratory analyses).

Manipulated variables (if applicable)

N/A

Prior knowledge of data (if applicable)

List the publications, conference presentations (papers, posters), and working papers (in prep, unpublished, preprints) you have worked on that are based on the data set. Describe which variables you have previously analyzed and which information you used in these analyses. Limit yourself to variables that are relevant to the current study. If the dataset is longitudinal, include information about which wave(s) of data were previously analyzed. Also, include any knowledge regarding missingness within the dataset or compliance, including at what level e.g. overall compliance, compliance for different types of reports, the mean level of compliance, range of compliance across participants.

None. However, during writing the pre-registration, the individual data of participants has been seen due to the individual data-viewing sessions that the participants were offered post data-collection. Moreover, data-cleaning has been done already and for this descriptive statistics have been calculated to check for out of range values. We also calculated scale scores and reliabilities for the baseline and follow-up Qualtrics questionnaires. Additionally, there are currently a few master-theses that use the across-time mean-aggregated ER Scales in relation to each other and to some Qualtrics questionnaires, such as self-esteem or neuroticism. Importantly, none of these worked with the raw time-series data!

What prior knowledge do you have at the time of pre-registration about trends in the data set you will be working with?

From data-cleaning, we know that there is an average compliance rate of 67% of the evening assessments (after removal of duplicate beeps and pilot-testers).

From the master theses (which have not been completed yet), we know that the *imean* and *iSD* (i.e., scores aggregated within individuals across time with the M and the SD) of the ER strategies are approximately normally distributed and positively correlated with each other on a between-person level. The between-person mean of *imean* is between 50 to 55 for the different strategies and the between-person mean of *iSD* is between 20 and 25.

No analyses have been conducted on the raw time-series data and within-person level.

Analysis Plan

Statistical models (required)

The data has a two-level structure: repeated measurements (daily, level 1) nested within individuals (level 2). We will conduct multilevel regression analyses.

Analyses for within-person relation between emotional characteristics and ER strategy use - linear and quadratic effects:

- Time-variant dependent variable: ER strategy (either single strategy (from 6 strategies in total) or ER strategy variability; interval)
- Time-variant independent variables: Linear and quadratic term of emotion characteristics (intensity and control; interval, person-mean centered)

- Time-variant covariates: Presence of micro-interventions (categorical, dummy-coded)
- Time-invariant covariates: Age (continuous, grand-mean centered) and Gender (categorical, dummy-coded) at baseline

For all analyses, we will include random intercepts and random slopes and their correlation for emotion characteristics (for the linear and quadratic terms). We will test for the significance of random effects by comparing nested models with increasingly random-effects structures using the Likelihood Ratio Test. The errors are assumed to be Gaussian distributed and serially correlated. The serial correlation of the errors will be modeled using an AR(1) process.

Separate models will be conducted for each model combination. For negative emotions, there are 14 models in total for the two different emotion characteristics (intensity and control) and the different ER strategy outcomes (six ER strategies: relaxation, expression, rumination, reappraisal, distraction, suppression or ER variability calculated with Bray-Curtis dissimilarity). For positive emotions, there are 2 models in total, because we only have one emotional characteristics variable (emotional intensity) and two ER strategies (expression, savoring).

What will you do should your data violate assumptions, your model not converge or some other analytic problem arises?

First, optimization algorithms will be adjusted with more iterations added.

Second, random effects will be removed in descending complexity, namely:

- correlation random slope quadratic term emotional characteristics and random intercept emotional characteristics
- random slope quadratic term emotional characteristics
- correlation random slope linear term emotional characteristics and random intercept emotional characteristics
- random slope linear term emotional characteristics

Transformations (if applicable)

The emotional characteristics and ER strategy use (scale 0-100) will be divided by 10 to prevent that the quadratic terms will get too large, which could potentially lead to problems with model estimation.

The time-varying predictors will be centered using the individual means (person-mean centered) and the time-invariant predictor will be centered using the grand mean.

Inference criteria (if applicable)

We will conduct two-tailed tests and use the inference criteria $p < 0.05$.

Missing data (if applicable)

We know the following information from data cleaning. For more information, please see the codebook <https://osf.io/fx3ay/files/osfstorage/62344cea980a08aa9516fa>:

6 participants dropped out during the study (i.e., 77 completed all 60 days), however we can use the data from the available days. Additionally, 1 person only received 1 evening assessment and two persons did not receive all evening assessments due to a technical error. The former person will be excluded, because there is no variation in the data across time.

79 of 83 participants have data on the baseline questionnaire (age, gender). Missing values on the baseline questionnaire will be imputed, so that people can be included in the multilevel models. Across all participants, 67% of all ESM questionnaires were filled in. The average number of filled in evening questionnaires was $M = 38$ per person ($SD = 15$, range = 1 - 60).

Data exclusion (if applicable)

Irrespective of whether data are pre-existing, will participants or observations be excluded based upon missingness, due to either technical issues or compliance? If so, specify a decision rule for this including:

How will you define a valid observation? We have no items that directly assessed careless responding. However, in accordance with earlier research (e.g., Dejonckheere et al., 2021), a zero variance of time-varying variables could be seen as careless responding. We will compute the variance for our primary predictors emotional intensity and emotional control and will exclude participants for whom this variance is zero. Additionally, measurement occasions where the entire questionnaire was completed in less than 1 minute will be excluded (the total questionnaire was 28 questions, including some open questions).

The following measurement occasions will be excluded, because they likely indicate technical errors (from data cleaning, we know that there are some occasions where this happened):

- there was a negative time between sending and opening the questionnaire (note: that from data cleaning, there were some cases with a negative time that were related to changing daylight saving times. Those cases will not be rated as invalid)
- the questionnaire was opened after it should have been expired (i.e., after 120 minutes)

How will you define compliance? Compliance will be defined based on the number of beeps responded to out of the total number of beeps that were delivered (which was 60 for most participants, but fewer for some participants with technical difficulties).

Which criteria will you use to define an outlier? How will outliers be handled? How are you going to deal with careless responding?

Outliers are defined as scores outside the 1% and 99% percentile, but we will not exclude participants based on outliers. See the data exclusions section for how we will deal with carelessness.

Exploratory analysis (if applicable)

To get more insight into the emotional characteristics, we will also conduct models, where we include an interaction between emotional intensity and control (for negative emotions only). We expect that higher intensity and lower control of negative emotions will be associated with more ER strategy use. However, because no research has been conducted on this topic, we did not make this a primary hypothesis.

Additionally, we will test the relationship between negative emotional intensity and momentary ER variability (i.e., the variability between all six negative ER strategies across time), as calculated with Bray-Curtis dissimilarity. Given that this index is new in the emotion regulation literature and because little research has been conducted in general on ER variability, we kept this research question exploratory.

Other

Other (if applicable)

Which software is used to estimate the statistical model(s)? Specify whether the default options or specific options of functions are used.

All analyses will be conducted in *R* (R Core Team, 2022). Multilevel models will be fit with the package *nlme* with the function *lme*. We use the ‘optim’ optimizer.

If there is any additional information that you feel needs to be included in your (pre-) registration, please enter it here, e.g. literature cited within your (pre-) registration.

Initially, we had the plan to also examine whether between-person differences in psychopathology would moderate the intra-individual covariation between emotional characteristics and ER strategy use. This is because the theoretical framework on ER flexibility suggests that individuals with lower psychopathology will show more context-sensitive ER strategy use (i.e., where the use of ER strategies systematically varies with the emotional characteristics/context in a beneficial direction), whereas individuals with higher psychopathology will either show no covariation between ER strategy use and emotional characteristics/context (i.e., context-insensitive ER strategy use) or an ER strategy use that covaries in an unexpected direction with the emotional characteristics/context [i.e., strategy-context misfit; Aldao et al. (2015), Bonanno & Burton (2013), Kalokerinos & Koval (in press)] However, simulation studies indicated that we did not have sufficient power for this cross-level interaction. For instance, power for the cross-level interaction between perceived stress and emotional control predicting reappraisal (a result that has been found earlier (Haines et al., 2016)) was 4%. As such, we decided to drop this research question for this study.

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